

Research article

The robust scheme for intrusion detection system in Internet of Things

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ABSTRACT

Machine learning and deep learning-based anomaly intrusion detection systems (IDSs) have become prevalent in securing IoT networks due to their ability to monitor traffic and detect zero-day attacks. However, recent studies highlight the high vulnerability of these models to adversarial attacks, in which minor input perturbations can significantly decrease the detection accuracy. Although many studies have focused on adversarial attack and defense techniques for deep learning, machine learning, particularly decision trees, has received limited attention. In this study, we aim to assess the efficacy of the robust decision tree in adversarial IoT environments. Our first experiments reveal the robust decision tree's sensitivity to the offset parameter. We thus propose a statistical approach to auto-select the offset value, enhancing model stability across varying attack offsets. Then, we present a robust scheme for IDSs in IoT environments. This approach employs the enhanced robust decision tree and a tabular deep learning model to detect and classify a range of cyber attacks. Our evaluation results on three popular IDS datasets—IoTID20, CIC-IDS-2017, and BOT-IoT—demonstrate that our proposed approach is robust under various adversarial attack conditions and achieves a consistent accuracy of over 95% in classifying different attack types.

1. Introduction

With recent advancements in machine learning, deep learning, and hardware technologies, Internet of Things (IoT) devices have been increasingly adopted in various industries [1]. These devices form a network of smart objects to provide various intelligent services, such as order tracking, system monitoring, predictive maintenance, and intelligent transportation [2]. For instance, deep reinforcement learning-based IoT was proposed to apply in early detection and diagnosis of lung cancer [3], which holds the potential for helping millions of patients. Another work at [4] presented an innovative approach to hardware design for IoT in addressing the problem of parking slot shortage in megacities. Lastly, the authors at [5] proposed to integrate a real-time network protocol like EtherCAT into automated guided vehicles.

However, the lack of security mechanisms for these devices poses a significant risk to information security and privacy. This deficiency is attributed to the constrained computational power and energy, which hinders the deployment of security mechanisms that require substantial computation, such as firewalls, anti-virus, and cryptography. Furthermore, IoT devices generate and transmit large amounts of data over the Internet, making them attractive targets for attackers. According to Sengupta et al. [6], denial of service/distributed denial of service (DoS/DDoS) and identity spoofing attacks are the most common attacks targeting IoT systems.

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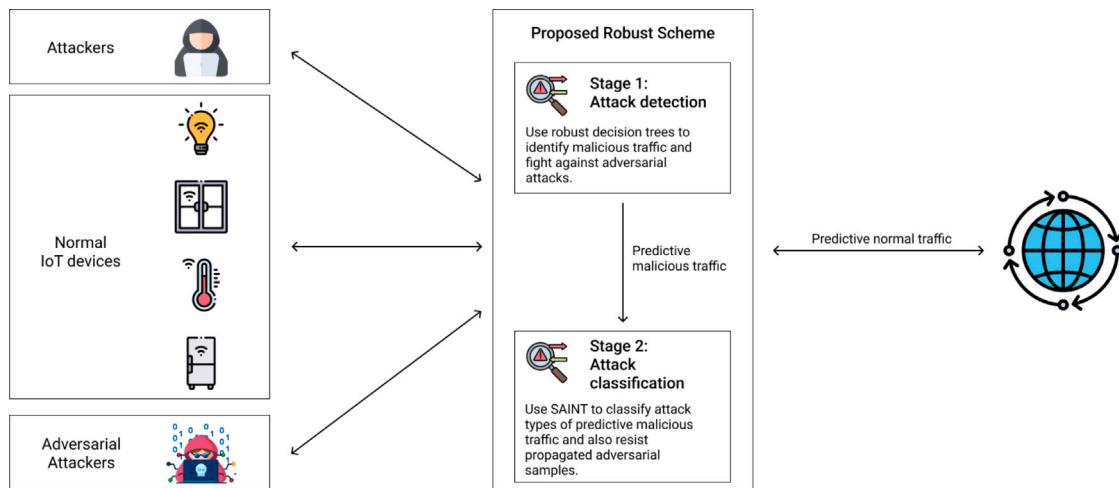


Fig. 1. Overview of the proposed robust scheme for intrusion detection system in IoT environments.

These attacks can compromise data or disrupt services, resulting in significant financial losses. Therefore, there is a critical need for robust and effective solutions to address security challenges in IoT devices.

This demand leads to significant efforts to secure IoT devices by offloading security mechanisms to other components within the IoT network, such as intrusion detection systems (IDSs) [7]. However, traditional signature-based IDSs are often limited in detecting sophisticated attacks. To address this limitation, machine learning, especially tree-based models, has been increasingly employed in IDSs to monitor traffic and detect suspicious behaviors across the entire network at low latency [8–10]. Furthermore, tree-based models have been shown to be more suitable for network traffic data, often encoded in tabular data, than deep learning models [11]. The authors in [11] have also proven that deep learning models tend to learn overly smooth functions and are more susceptible to irrelevant features. Meanwhile, tree-based models can learn irregular patterns and are more resilient to non-informative features in tabular data. Therefore, the deployment of tree-based IDS models in IoT networks could improve the security and privacy of IoT networks.

Despite the various advancements in IDSs using machine learning, particularly tree-based models, they still encounter a significant threat from adversarial attacks [12,13]. These attacks are designed to evade the detection of IDS by modifying the network payloads without the administrators' knowledge [14]. In detail, the typical work by Papernot et al. [15] demonstrated the efficacy of their practical black-box adversarial attacks by successfully causing misclassifications in machine-learning models used by Amazon and Google. The misclassification rates on the models of these two platforms are 96.19% and 88.94%, indicating severe impacts of adversarial attacks on the model performance. Therefore, it raises the need for defense techniques against such adversarial attacks to protect IoT networks and devices.

The motivation for this article spurs from the following basic assumption: It is challenging to defend against adversarial attacks with an unknown attack offset that target tree-based IDS in the IoT network. We gave shape to this assumption after reading several research papers [16–19] in this field where, in essence, the authors focus only on adversarial attacks and defenses on deep learning models, while the works on decision trees are limited. Furthermore, the work in [20] and the improved version in [21] improve the robustness of decision trees; however, our experiments showed that the robustness of their models is susceptible to parameters (i.e., attack offsets). The defense offset used to train a robust decision tree must be set appropriately to the attack offset used by adversarial attackers, which is unknown in practical scenarios. Therefore, we evaluate the sensitivity of their robust decision tree and propose a solution to minimize this sensitivity in this work. By doing so, the decision tree becomes more robust under adversarial attacks.

The main idea of this paper is to build a robust and high-performance IDS for IoT networks. The proposed IDS could effectively defend against adversarial attacks without knowing the attack offset values and accurately detect different attack types with low latency. To achieve the first objective, we adopt a statistical approach to dynamically calculate the defense offset, rather than setting a fixed value in advance. In particular, by leveraging the 68-95-99.7 rule, we employ three standard deviations to determine the defense offset, which could dynamically scale according to the features of different datasets. For the second objective, we design a detection scheme that comprises two distinct stages. The first stage employs a robust decision tree to detect attacks, while the second stage using a deep-learning model, is responsible for attack classification based on the outcomes from the first stage. Any abnormal network traffic that the first stage identifies is rerouted to the second stage for further attack classification. The overview of our scheme is illustrated in Fig. 1. We assume that the newly improved decision tree is now resilient to attack offsets, but a crucial issue remains concerning the ability of the second detection stage to withstand “propagated” adversarial samples. In our scheme, as the first-stage model is directly exposed to end devices, we assume that evasion attacks mainly target this first-stage model. Thus, adversarial attacks targeting the first detection stage introduce noises to the samples, making resilience against noisy data necessary

for the second detection stage to classify attacks accurately. For our second-stage model, we adopted SAINT, a deep learning model designed explicitly for tabular data and claimed to be resilient to missing and noisy data [22]. In summary, our main contributions are:

- Assess the parameter sensitivity of the current state-of-the-art robust decision tree, as described in the paper [20].
- Present a method to improve the robust decision tree to mitigate its parameter sensitivity, thereby increasing the model's resilience to various adversarial attack offsets.
- Assess the efficacy of SAINT, a deep learning model tailored explicitly for tabular data, on detecting distinct attack categories in diverse adversarial scenarios.
- Propose a robust IDS scheme consisting of two stages (i.e., attack detection and attack classification) resistant to adversarial attacks.

This paper is structured as follows: Firstly, we delve into various related works concerning machine learning- and deep learning-based IDS, as well as various adversarial attack and defense techniques in Section 2. Then, we provide details about our threat model in Section 3. Next, we outline our robust scheme's architecture and operational steps in Section 4. Finally, we present the experimental results of our proposal in terms of detection performance and robustness in Section 5.

2. Related works

2.1. Machine learning- and deep learning-based IDS

Numerous studies have investigated anomaly-based IDSs employing machine learning and deep learning techniques, utilizing various algorithms, architectures, and methodologies [23,24]. A comprehensive review conducted by Thakkar et al. [25] examined several aspects of implementing IDSs for IoT environments, including IDS placement strategy, IDS analysis strategy, types of intrusions, and attack detection techniques. They also summarized 24 works employing machine learning and 20 others using deep learning for intrusion detection in IoT networks between 2015 and 2019. Another systematic review was presented at [26], where the authors surveyed a total of 324 papers and conducted an in-depth analysis of over 40 main studies regarding IDSs in IoT. Based on the analysis, they proposed a detailed categorization of 4 detection approaches, 2 evaluation methodologies, and 9 attack types. Furthermore, a survey by Ring et al. [27] focused on labeled datasets for evaluating machine learning-based IDSs, grouping existing datasets into flow-based and packet-based network data. Within each dataset group, they thoroughly examined 15 different properties of individual datasets to identify their unique characteristics and provided recommendations on which datasets to use in different scenarios.

In addition to literature reviews and surveys, research efforts have been dedicated to enhancing the detection capabilities of IDSs. For example, in our previous work [8], we introduced MidSiot, a multistage IDS designed for IoT networks that utilize decision trees with the CART algorithm to profile device types, detect abnormal traffic, and classify various network attack types. Our experiments on three different datasets, IoTID20 [28], CIC-IDS-2017 [29], and BOT-IoT [30], showed that MidSiot achieved an average accuracy of 99.98% and 99.68% for attack detection and classification, respectively, while also meeting the requirement of low latency. Similarly, Anthi et al. [31] proposed a hierarchical IDS structure that utilizes decision trees (with the J48 algorithm), achieving a high F1-score of over 97% for attack detection and classification. However, experiments on unseen data revealed a significant drop in accuracy from 97% to 90%.

Other studies have also explored deep learning-based methods for intrusion detection. For instance, [32] proposed a novel neighborhood search-based particle swarm optimization algorithm to train seven deep convolutional neural network architectures for their IDS. Their methodology achieved an average accuracy of 96.40% on the UNSW-NB15 and BOT-IoT datasets. Another deep learning-based IDS, Realguard, was proposed in [33]. It was designed to be low-cost and capable of real-time operation. The authors used a neural network with gradually increasing layers and neurons until their performance criteria were met, in which their final model has five hidden layers and 34,315 parameters. Their experiment was conducted on the CIC-IDS-2017 dataset, and the model was deployed on the Raspberry Pi 4B, showing that their IDS achieved an average accuracy of 99.57% and a false positive rate of 0.4%, with also a high packet processing rate of about 10,600 packets per second. In [34], the authors proposed a hybrid feature selection method that combined particle swarm optimization, ant colony algorithm, and generic algorithm with two-stage classifier ensembles. By testing their approach on the NSL-KDD and UNSW-NB15 datasets, they achieved a high accuracy of 85.8%, outperforming other classification techniques recently proposed and tested on these two datasets.

Further on applying machine learning and deep learning models in IDS, researchers are currently exploring the potential of deep learning on tabular data. While images, audio, and texts possess a specific feature ordering, and the features of each sample are of the same form, tabular data exhibits more heterogeneity in these aspects. They have made tree-based classifiers superior in tabular data, given their built-in feature selection, the ability to learn the target irregular function, and non-rotational invariance characteristics [11]. However, researchers have proposed tabular deep learning architectures to overcome these characteristics and unveil the spatial relationships in data. One such architecture is TabNet, as proposed by Arik [35]. TabNet mimics the feature selection mechanism of tree-based classifiers by utilizing a learnable feature mask, allowing the model to choose the most important features for each sample at every decision step. The gathered information is aggregated to highlight the most essential features of the dataset as a whole. Experimental results showed that TabNet outperformed state-of-the-art gradient-boosted trees while also providing interpretability. Another architecture, TabTransformer, proposed by Huang [36], leverages self-attention-based transformers to embed categorical data contextually. TabTransformer was demonstrated to achieve high accuracy and robustness

against noisy and missing data features. The most recent and state-of-the-art tabular deep learning model, SAINT, was introduced by Somepalli [22]. SAINT outperformed TabNet and TabTransformer by contextually embedding dense numerical and categorical features as well as introducing a new inter-sample attention layer. Experimental results demonstrate that SAINT outperforms other state-of-the-art models, including XGBoost, TabNet, and TabTransformer, regarding the mean AUROC score. Considering the advances of SAINT, we utilize it in our attack classification task to achieve high accuracy and robustness against noisy data produced by adversarial attacks.

2.2. Defenses against adversarial attacks

Deep learning has attracted substantial attention from academia and industry due to its remarkable ability to solve complex problems efficiently. Consequently, researchers have begun examining the intrinsic properties of deep learning models to identify potential vulnerabilities. Szegedy et al. [37] conducted one of the pioneering works in this area and successfully identified two intriguing properties of neural networks. They demonstrated that deep neural networks learn discontinuous input–output mappings that can be exploited to cause the network to misclassify an image by inserting an imperceptible perturbation. Additionally, they found that the same perturbation applied to the same input can be used to attack a different network trained on a different subset of the original data. Building on this work, Cubuk et al. [38] further explored the intriguing properties of adversarial samples. They argued that these samples are not generated due to the high dimensionality or linearity of the data but rather due to the inherent uncertainty of neural networks in their predictions. The authors also examined the effect of different network architectures on adversarial sensitivity and used reinforcement learning to identify adversarially robust architectures.

As the interest in adversarial attacks extends beyond computer vision, researchers have presented various attack and defense techniques for deep learning models in different fields, such as natural language processing, information security, and even the physical world [39,40]. In the realm of information security and intrusion detection, much research has been devoted to evading IDSs [41]. One approach to generating adversarial samples involves using Generative Adversarial Networks (GANs), which comprise three components: a generator, a discriminator, and a virtual IDS. In [42], the authors used GANs to subtly transform regular attack traffic into an adversarial form. The discriminator was employed to differentiate between the generated traffic and natural traffic. However, the authors mistakenly used the discriminator to simulate a black-box IDS, causing their proposed system to fail to preserve the original attack network traffic functionality. In contrast, other proposed models, such as those in [41,43], also used GANs to generate adversarial samples but succeeded in preserving the original network traffic functionality while ensuring a high attack success rate.

Furthermore, various attack techniques and defense strategies have been extensively researched [44,45]. The most common and effective attack technique is still gradient descent over the loss surface of visual models, as highlighted in the survey by [45]. While many defense strategies have been proposed, adversarial training remains the most promising approach. According to the authors in [45] claimed that other strategies mostly rely on ad-hoc rules and heuristics, which are susceptible to stronger attacks or different attack conditions. Adversarial training is a technique in which we explicitly train a model on a dataset of clean and adversarial samples, making it more robust against adversarial samples and less susceptible to noise in clean data [46]. Various defense methods have been introduced to improve the initial adversarial training [46]. However, despite these extensive efforts, adversarial training alone has only partially resolved adversarial attacks. As [46] points out, other problems remain unresolved, particularly min–max optimization and overfitting, which can cause adversarially trained deep learning models to remain vulnerable to such attacks.

Although deep learning models have gained traction in the research community over the years regarding adversarial attacks and defenses, only a few works have been done on tree-based models. The first publication to mention adversarial attacks against decision trees was [47]. In this work, the authors introduced the concept of *adversarial sample transferability*, where adversarial samples created to attack one model could also be used to attack a different model regardless of their learning algorithms, parameters, and datasets. To create an adversarial sample against a decision tree, one can search for a neighboring leaf with a different class than the original sample and adjust the feature values of the original sample accordingly. However, Chen et al. [20] have proposed an improved training algorithm for decision trees that can resist adversarial samples created with the above technique. This new algorithm maximizes information gain in the optimal split search under the adversarial attacker's capabilities. Decision trees trained with this new algorithm exhibit good robustness, with their accuracy under adversarial attacks being competitive with regular trees. However, this algorithm is time-consuming, and therefore, the authors in [21] have improved the algorithm to reduce the time complexity from $O(N)$ to $O(1)$. Our experiments with this algorithm have shown that although the robust decision trees could achieve good accuracy and robustness against adversarial attacks, they are still sensitive to parameters, especially the offset used for training (i.e., the defense offset). Therefore, in this work, we proposed an enhancement to this algorithm that reduces sensitivity to this parameter while maintaining robustness and accuracy.

3. Threat models

In this study, we examine the existence of both typical and adversarial network attackers in the local IoT network. The objective of the typical attackers is to attack other devices within the network, and other targets on the Internet, and adversarial attackers adopt adversarial techniques to bypass our attack detector.

Our research considers a wide range of network attacks, including DoS/DDoS, botnet, man-in-the-middle attacks, scanning (host, port, and OS), brute-force attacks, infiltration, web attacks, and theft (keylogging and data exfiltration). Additionally, we focus on

Table 1
Notations and definitions.

Notation	Definition
ϵ	The adversarial offset
η	The temporary splitting point
X	The training set
$X_{(:,j)}$	The list of the j th feature's values of all samples in X
X^{vuln}	The set of perturbable samples
X_{η}^{worst}	The set of perturbable samples leading to the worst information gain w.r.t η
x_i	The i th sample of X
y_i	The ground truth of x_i
x_i^{adv}	The adversarial sample corresponding to x_i
$x_{(i,j)}$	The j th feature's value of x_i
$x_{(i,j)}^{adv}$	The j th feature's value of x_i^{adv}
p_i	The decision path of x_i
p_i^{adv}	The decision path of x_i^{adv}
l_i	The leaf node w.r.t x_i
l_i^{adv}	The leaf node w.r.t x_i^{adv}
$f(x_i)$	The decision tree function to make an inference on the sample x_i
$f_{path}(x_i)$	The function to get the decision path of the sample x_i
$f_{leaf}(x_i)$	The function to get the leaf node of the sample x_i
$f_{class}(l_i)$	The function to get the class of the leaf node l_i
$f_{neighbor}(l_i)$	The function to find a neighbor leaf of the leaf l_i
$f_{diff}(p_1, p_2)$	The function to get the perturbable features according to the two decision paths p_1 and p_2
$IG(X)$	The function to calculate the information gain of X

adversarial attacks and assume that adversarial attackers possess knowledge of the decision trees' internal structure, which could be obtained through a model extraction attack targeting decision trees [48]. Upon learning the tree structure, attackers can execute a decision tree evasion attack, as proposed in detail in Algorithm 1. While various adversarial attack techniques could be employed against a deep learning model and then transferred to decision trees, the impact demonstrated in Papernot's work [47] suggests that specialized attack techniques aimed at decision trees are more effective than other transfer-based attack techniques. Therefore, we only employ this simple and effective attack technique to assess the robustness of our decision tree.

The Algorithm 1 provides an overview of the adversarial attack technique. The attacker's initial step is to traverse the decision tree to construct a decision path that corresponds to a given non-adversarial sample. If the prediction of the tree is correct, the attacker proceeds to perturb the sample to evade the tree; otherwise, the sample is ignored. In the perturbation step, the attacker searches for a neighboring leaf that belongs to a different class. Once a suitable neighboring leaf is identified, the features of the non-adversarial sample are perturbed with respect to the new decision path. It is worth noting that in a real-world scenario, an attacker might only perturb malicious traffic; however, our work does not limit this capability, and therefore both normal and abnormal traffic can be perturbed.

In Algorithm 1, we introduce the concept of the *offset* ϵ , representing the constraint for adversarial perturbations. This constraint defines the L_{∞} norm bound of the perturbations, ensuring that no individual feature is perturbed by more than ϵ . For example, given a feature j , a splitting point η , an offset ϵ , and a sample x_i , if $x_{(i,j)}$ falls within the range of $[\eta - \epsilon, \eta + \epsilon]$, it is considered perturbable and vulnerable to adversarial attacks with the offset ϵ . On the other hand, if $x_{(i,j)}$ is outside of this range, modifying the sample by an amount of ϵ would not change its predictive class. Therefore, such unperturbable samples are ignored by the algorithm. Detailed definitions of notations used in this paper are provided in Table 1.

4. Proposed IDS

4.1. The system overview

The proposed scheme for IDSs in IoT environments comprises two main stages: attack detection and attack classification. The first stage aims to identify malicious traffic, while the second stage is responsible for classifying attack types. To achieve effective attack detection at the first stage, we suggest deploying the IDS on a local IoT gateway, allowing it to directly monitor network traffic within the local network and quickly detect attacks from such traffic. In contrast, the second stage could be deployed on a cloud platform or an ISP gateway with sufficient computational power. At this stage, we use a single universal classifier to identify the types of attacks from all predictive malicious traffic. This approach reduces the computational overhead on the local IoT gateway and provides a centralized platform for analyzing and identifying attack types across different devices.

As illustrated in Fig. 2, our scheme consists of five operational modules. The first module captures inbound and outbound raw packets and transmits them to a network flow generator. This generator uses deep packet inspection to consolidate packets into flows that share common source/destination IP addresses, ports, and protocols. It also calculates additional flow features and statistics. Identifiable features of the network flow, such as IP addresses, MAC addresses, and flow IDs, are removed before flows are fed into the attack detector. The detector is powered by the robust decision tree designed to resist adversarial attacks while ensuring high detection accuracy, as discussed in Section 4.2. Suppose that the first detection stage detects an abnormal network flow; it is

Algorithm 1 The algorithm to craft an adversarial sample against a decision tree.

Input: a decision tree $f(\cdot)$

Input: a sample (x_i, y_i) , $x_i \in [0, 1]^d$, $y_i \in \{0, 1\}$

Input: an offset ϵ

Output: an adversarial sample x_i^{adv}

```

1: if  $f(x_i) \neq y_i$  then
2:   return  $x_i$ 
3: end if
4:  $p_i \leftarrow f_{path}(x_i)$ 
5:  $l_i \leftarrow f_{leaf}(x_i)$ 
6:  $l_i^{adv} \leftarrow l_i$ 
7: while  $f_{class}(l_i^{adv}) \neq f_{class}(l_i)$  do
8:    $l_i^{adv} \leftarrow f_{neighbor}(l_i)$ 
9: end while
10:  $p_i^{adv} \leftarrow f_{path}(l_i^{adv})$ 
11:  $J_i^{adv} \leftarrow f_{diff}(p_i, p_i^{adv})$ 
12: for  $j \in J_i^{adv}$  do
13:    $x_{(i,j)}^{adv} \leftarrow x_{(i,j)}^{adv} + k$  such that  $k \in [-\epsilon, \epsilon]$ 
14: end for
15: return  $x_i^{adv}$ 

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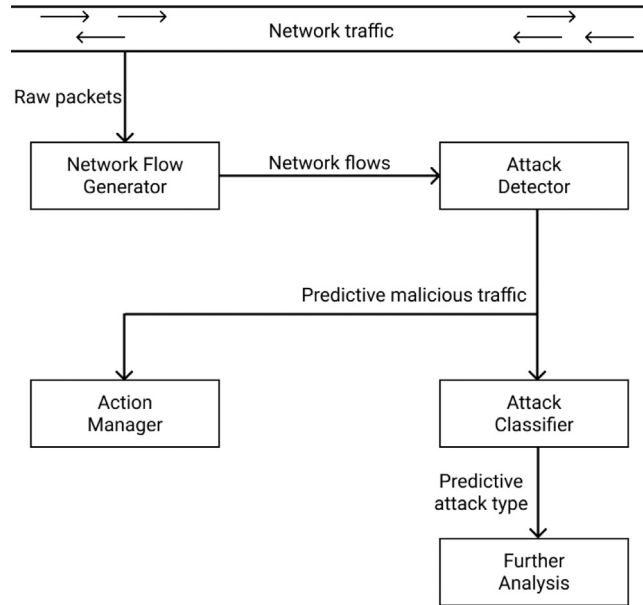


Fig. 2. Block diagram of the operation of the proposed robust scheme.

forwarded to the action manager, which generates alerts about the potential attack and forwards the flow to the second detection stage for attack classification. To ensure robustness against noisy data and adversarial attacks, we employ a state-of-the-art tabular deep learning model for attack classification, which is well-explained in Section 4.3.

4.2. Robust attack detection

4.2.1. Robust decision tree

We describe the training procedure for the robust decision tree in Algorithm 2. Typically, the algorithm identifies a feature and value that maximizes the detection accuracy and robustness against adversarial attacks. In detail, the training procedure starts by traversing over each feature j in the feature set d of the training data X . The algorithm then iterates over each value η of the feature j and calculates the information gain. To achieve robustness, the algorithm considers the condition of an adversarial attack with a

Algorithm 2 The algorithm to find a robust split.

Input: the training set $X = \{(x_i, y_i)\}_{i=1}^N$, $x_i \in [0, 1]^d$, $y_i \in \{0, 1\}$.

Input: the offset ϵ .

Output: the optimal split η_{best} of the current node.

```

1: for  $j \in [1, d]$  do
2:   for  $\eta \in \text{sorted}(X_{(:,j)})$  do
3:      $X^{vuln} \leftarrow \{(x_i, y_i) | x_{(i,j)} \in [\eta - \epsilon, \eta + \epsilon]\}$ 
4:      $X_{\eta}^{worst} \leftarrow \arg \min_{X^{vuln}} IG(X)$ 
5:   end for
6: end for
7:  $\eta_{best} \leftarrow \arg \max_{X_{\eta}^{worst}} IG(X)$ 
8: return  $\eta_{best}$ 

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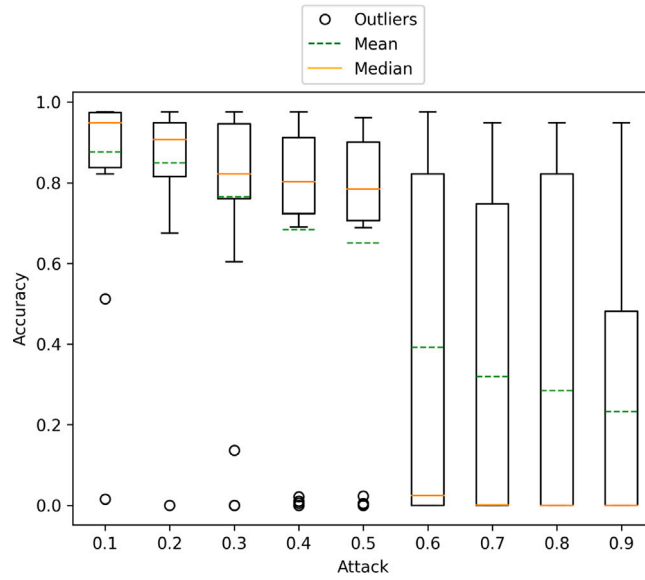


Fig. 3. Comparison of attack detection accuracy of robust decision trees under different attack offsets on the IoTID20 dataset.

given offset ϵ . The feature value with the highest possible information gain under ϵ becomes the optimal split. The algorithm then recursively proceeds to the child nodes until a stopping criterion is satisfied.

Although this training procedure for robust decision trees does not require adversarial samples, it is still more computationally expensive than the traditional decision tree. In particular, Algorithm 2 tries to find a split that maximizes the minimum information gain obtained by all possible perturbations within the attack offset ϵ . Consequently, at each feature value η , the algorithm needs to perform all possible perturbations for all vulnerable samples within ϵ distance to η in order to find the minimum information gain. Despite this computational overhead, it is an effective defense strategy, ensuring that the decision tree is accurate and robust against adversarial attacks that directly target this model.

4.2.2. The problem of parameter sensitivity

The sensitivity to the defense offset parameter of the robust decision tree is evaluated through experiments with various attack and defense offsets. The evaluation is conducted on three well-known IDS datasets: IoTID20, CIC-IDS-2017, and BOT-IoT.

The distribution of attack detection accuracy for the robust decision trees in nine attack offsets is depicted in Figs. 3, 4, and 5. The green dashed lines in these boxes represent the mean accuracy, while the orange lines represent the median. Additionally, outliers are represented by circles.

The results indicate that robust decision trees exhibit reduced vulnerability to adversarial attacks with smaller offsets. However, as the attack offset increases, the accuracies decline and become more inconsistent. This observation underscores the importance of selecting an appropriate defense offset while training the model. Moreover, it underscores the need for an algorithm to determine the optimal defense offset in real-world scenarios automatically.

In our examination of the IoTID20 dataset results, we observed that robust decision trees could maintain a decent level of accuracy when dealing with minor attack offsets, as illustrated in Fig. 3. However, the accuracy distribution displays considerable variability as the attack offset increases. Specifically, for attack offsets within the range of 0.1 to 0.5, the average accuracy peaks

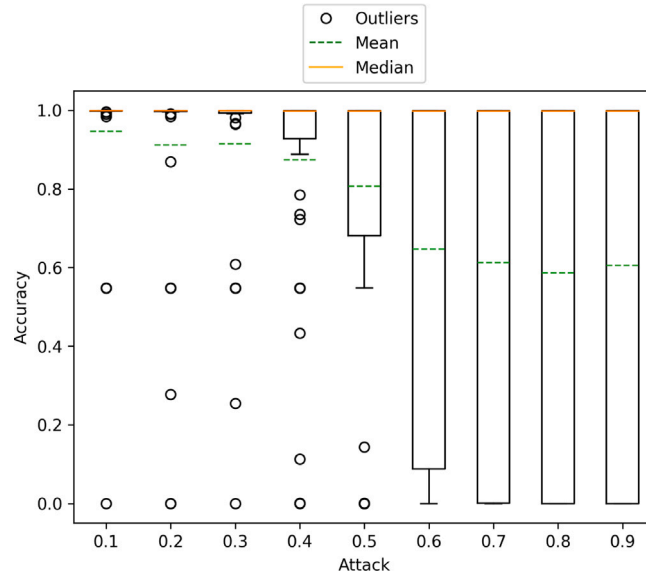


Fig. 4. Comparison of attack detection accuracy of robust decision trees under different attack offsets on the CIC-IDS-2017 dataset.

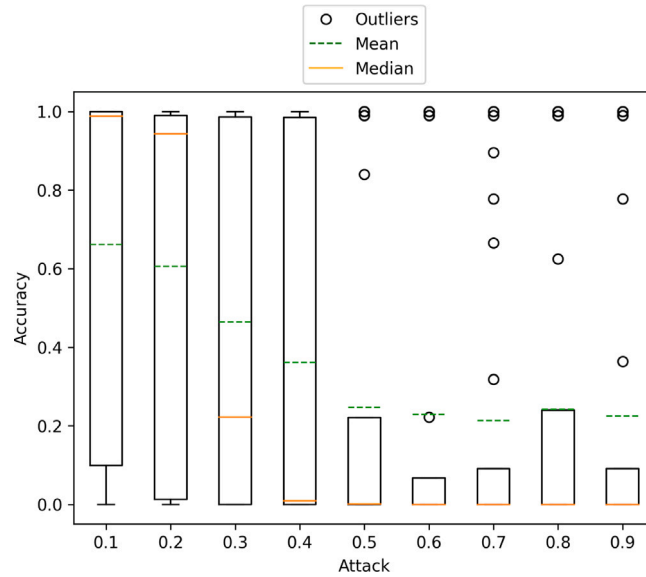


Fig. 5. Comparison of attack detection accuracy of robust decision trees under different attack offsets on the BOT-IoT dataset.

at 88%, with an interquartile range (IQR) of less than 20%. Conversely, when the attack offset reaches 0.6, the IQR dramatically expands to 82%, and the mean accuracy decreases by half. This marginal IQR indicates high unpredictability in attack detection accuracy at this offset.

Regarding the CIC-IDS-2017 dataset, as illustrated in Fig. 4, the results reveal that robust decision trees exhibit decent and surprisingly stable mean accuracy levels when confronted with small attack offsets. However, their ability to maintain detection stability diminishes significantly as attack offsets increase. Specifically, for attack offsets ranging from 0.1 to 0.3, the mean accuracy remains high, fluctuating between 91% to 95%, with an IQR that is nearly negligible. Nevertheless, as the attack offsets increase, the mean accuracy steadily declines, reaching 81% at an attack offset of 0.5, and plunges to a mere 59% at an attack offset of 0.8. Notably, under these high attack offsets, the IQR predominantly spans the entire range of accuracy values, indicating extreme instability in the performance of robust decision trees.

The experimental results from the BOT-IoT dataset, depicted in Fig. 5, indicate that attack detection accuracy is at its lowest and most unpredictable among the three datasets analyzed. Specifically, the highest mean accuracy observed is a mere 66% at an attack offset of 0.1, which then marginally declines to 25% at an attack offset of 0.5. Moreover, the IQR for attack offsets ranging from

0.1 to 0.4 is substantial, revolving around 99%. As the attack offsets increase, the detection accuracy further deteriorates, with the highest recorded accuracy reaching only 25%. However, it is worth noting that the IQR for these higher attack offsets is relatively low, averaging around 14%.

4.2.3. A solution to parameter sensitivity

We aim to improve the resilience of decision trees against the offset parameter by proposing a statistical approach to calculate the defense offset ϵ dynamically rather than setting a fixed value in advance. Details of this solution are provided in Algorithm 3. Specifically, instead of obtaining the defense offset ϵ from user input, Algorithm 3 calculates the three standard deviations of the feature values and uses it as the ϵ . In statistical theory, the standard deviation measures how far each value in a dataset deviates from the mean value. The 68-95-99.7 rule demonstrates that about 68% of all the data falls within one standard deviation of the mean, approximately 95% falls within two standard deviations, and around 99.7% falls within three standard deviations.

To calculate the standard deviation, we take the square root of the variance. Additionally, the variance is computed by taking the squared differences between each value and the population's mean, then dividing this sum by the number of values in the population. In mathematical terms, the variance is represented by Eq. (1), where X is the population or set of random values, X_i is the i th value, μ is the mean, and n is the total number of values in the population. Similarly, the standard deviation is represented by Eq. (2).

By dynamically calculating the defense offset ϵ that aligns with the data distribution of each feature, our proposed approach can adapt to the characteristics of the features and provide a more robust defense against adversarial attacks. The empirical results presented in Section 5 provide evidence that using three standard deviations as the new offset can effectively reduce the sensitivity of the robust decision tree to the offset parameter.

Algorithm 3 The algorithm to find a robust split.

Input: the training set $X = \{(x_i, y_i)\}_{i=1}^N$, $x_i \in [0, 1]^d$, $y_i \in \{0, 1\}$.

Output: optimal split for the current node.

```

1: for  $j \leftarrow 1$  to  $d$  do
2:    $\epsilon \leftarrow 3 * std\_dev(X[j])$ 
3:   for  $\eta$  in sorted( $X[j]$ ) do
4:      $vuln\_set \leftarrow \{(x_i, y_i) | x_i[j] \in [\eta - \epsilon, \eta + \epsilon]\}$ 
5:      $worst\_cases[j][\eta] \leftarrow \arg \min_{vuln\_set} IG(X)$ 
6:   end for
7: end for
8:  $best\_split \leftarrow \arg \max_{worst\_cases} IG(X)$ 
9: return  $best\_split$ 

```

$$var(X) = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2 \quad (1)$$

$$std_dev(X) = \sqrt{var(X)} \quad (2)$$

4.3. Robust attack classification

In the subsequent detection stage of our proposed scheme, we aim to classify normal malicious and adversarially perturbed traffic. However, this poses a challenge as the second detection stage might be susceptible to noise from the perturbations of the adversarially malicious traffic. To address this issue, we employed a state-of-the-art tabular deep learning model known as SAINT [22]. This model has been custom-designed to yield optimal performance on tabular data while being resilient to noisy data. By leveraging the capabilities of this sophisticated model, we can effectively identify and distinguish between different types of attacks while maintaining stability and robustness against adversarial samples propagated from the first detection stage.

In Fig. 6, we provide an overview of the architecture of SAINT. The model comprises five key components: the input layer, embedding layer, SAINT, Multilayer Perceptron (MLP), and output layer. As an initial step, the input layer of the model receives a d -dimensional vector, $x_i = [x_i^1, x_i^2, \dots, x_i^d]$, which corresponds to the i th data sample. To enhance the representation of the input, a special token, [CLS], is appended to the vector. This token, derived from the BERT model [49], encapsulates the entire input and provides contextual information about its overall meaning. By incorporating this representation, downstream models can better learn the input and make more precise predictions.

In contrast to the TabTransformer model, which only contextually encodes categorical features and then concatenates continuous features to the embedded categorical features, the SAINT's embedding layer addresses this limitation by projecting both continuous and categorical features into a higher dimensional embedding space. In the case of TabTransformer, continuous features do not undergo any attention block, which leads to the model's inability to relate these features to one another or categorical features. In SAINT, all features are subjected to transformer blocks, and different embedding functions can be employed for different features. By contextually embedding continuous and categorical features, SAINT can better capture the interdependencies among all features and improve its predictive performance.

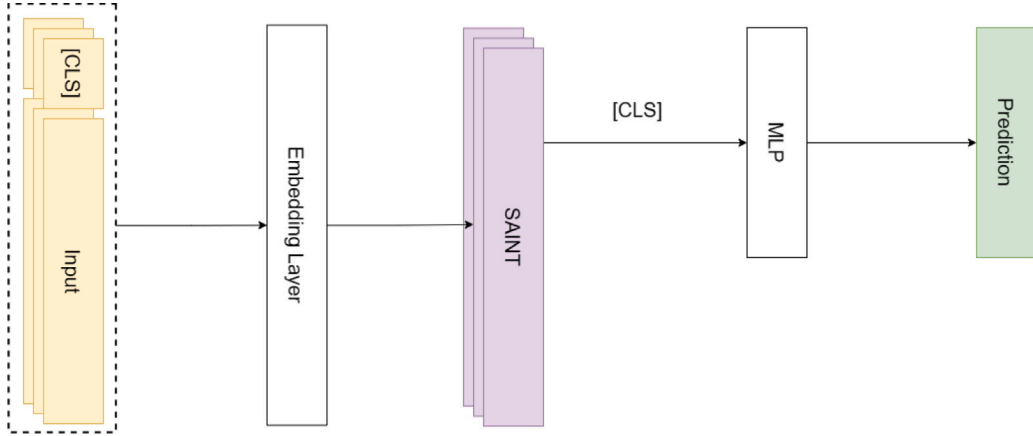


Fig. 6. The overall architecture of SAINT.

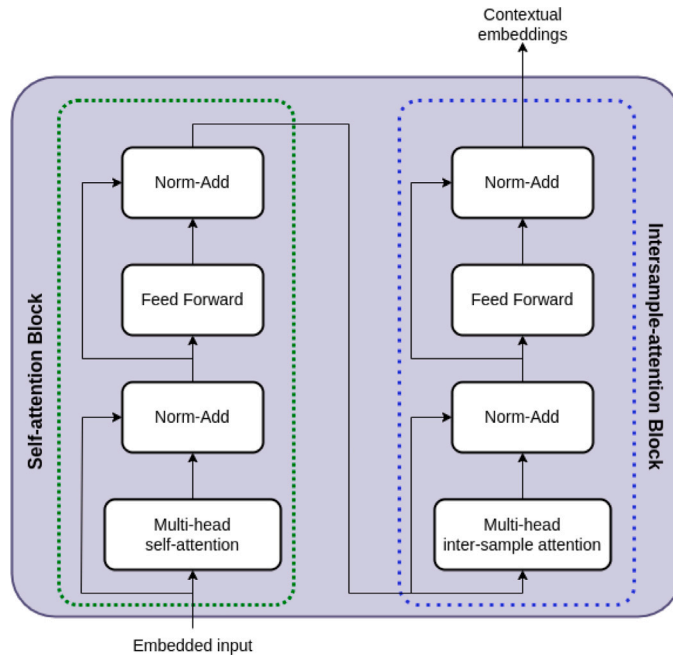


Fig. 7. The inner structure of the SAINT sub-layers.

The SAINT layer, a vital component of the SAINT architecture, comprises L identical sub-layers, each containing one self-attention transformer block and one intersample attention transformer block. The structure of a SAINT sub-layer is depicted in Fig. 7, and the mathematical demonstration is provided in Eq. (3). In more detail, the self-attention encoder (SA) learns the relationships among feature embeddings of the augmented input vector to create contextual representations of the same dimension. The output representations are then regularized using a Norm-Add ($Norm$) layer, which restricts their values from becoming too large or too small and avoids over-fitting. Additionally, a residual connection from the embedding input to the output of the Norm-Add layer is used to prevent the gradient from vanishing [50]. The Feed Forward (FF) layer, which also receives a sequence of embeddings, is passed through a Norm-Add layer and incorporates a residual connection. The intersample attention block is similar to the self-attention block, except for the first step, where the intersample attention encoder (IA) replaces the self-attention encoder. The outcome of this procedure, b_i^4 , represents the contextual embeddings of the data point x_i .

$$\begin{aligned}
 b_i^1 &= Norm(SA(E(x_i))) + E(x_i) \\
 b_i^2 &= Norm(FF(b_i^1)) + b_i^1 \\
 b_i^3 &= Norm(IA(b_i^2)) + b_i^2 \\
 b_i^4 &= Norm(FF(b_i^3)) + b_i^3
 \end{aligned} \tag{3}$$

Table 2
Devices, attack categories, and subcategories of the IoTID20 dataset.

Device	Category	Subcategory
EZVIZ, NUGU, External	Normal, DoS, Mirai, MITM, Scan	Normal, Syn Flooding, Brute Force, HTTP Flooding, UDP Flooding, ARP Spoofing, Host Port, OS

Table 3
Devices, attack categories, and subcategories of the CIC-IDS-2017 dataset.

Device	Category	Subcategory
Web server 16 Public, Ubuntu server 12 Public, Ubuntu 14.4 32 bit, Ubuntu 14.4 64 bit, Ubuntu 16.4 32 bit, Ubuntu 16.4 64 bit, Win 7 Pro, Win 8.1 64 bit, Win Vista 64 bit, Win 10 pro 32 bit, win 10 64 bit, MAC, External	Benign, Bot, Brute Force, Dos/Ddos, Infiltration, Portscan, Web Attack	Benign, Bot, FTP-Patator, SSH-Patator, DDoS, DoS, GoldenEye, DoS Hulk, DoS Slow, httpstest, DoS slowloris, Infiltration, Portscan, Web Attack-Brute Force, Web Attack-Sql Injection, Web Attack-XSS, Heartbleed

Table 4
Devices, attack categories, and subcategories of the BOT-IoT dataset.

Device	Category	Subcategory
Ubuntu server, Ubuntu mobile, Metasploitable, Windows 7, Ubuntu tap, External	Normal, DoS/DDoS, Reconnaissance, Theft	Normal, Service scanning, OS Fingerprinting, DoS/DDoS TCP, DoS/DDoS UDP, DoS/DDoS HTTP, Keylogging, Data Exfiltration

Regarding intersample attention, the SAINT model uses this type of attention to enable cross-row attention in tabular data. Unlike self-attention, which focuses only on the features of a row, intersample attention computes attention across multiple rows from the feature embeddings of each row. This approach allows SAINT to improve the representation of a single data point by inspecting other similar points. Moreover, when missing or noisy data appears in one row, the intersample attention transformer allows SAINT to borrow the corresponding features from other rows. As a result, this intersample attention makes the model less prone to noisy or missing data and capable of learning relationships among samples.

At the end of the SAINT layer, it generates the contextual embeddings of a batch of input, enabling downstream models to make accurate predictions based on the learned representation. In particular, each sample in the input layer is appended with a [CLS] token that represents the contextual information of the overall meaning of the input. Each token encodes the overall knowledge learned from the self-attention and intersample attention of a single data point. Therefore, the [CLS] token can be used by a downstream model (such as MLP) to perform classification. In our context, the MLP leverages the information encapsulated in the [CLS] token to classify different attack types.

5. Evaluation results

5.1. Datasets and evaluation metrics

5.1.1. Datasets

To assess the efficacy and resilience of our proposed scheme, we utilized three widely recognized datasets: IoTID20, CIC-IDS-2017, and BOT-IoT. The IoTID20 dataset encompasses traffic data from two devices and four distinct attack categories. The CIC-IDS-2017 dataset, on the other hand, includes network traffic information from twelve devices, each labeled based on their architecture and operating system, and features six diverse attack categories. Lastly, the BOT-IoT dataset comprises network traffic from five devices and three types of attacks. Further elaboration regarding the devices and attack types for these three datasets can be found in Tables 2, 3, and 4, respectively. Additionally, all dataset features were normalized to the range of [0, 1] for ease of evaluation.

5.1.2. Evaluation metrics

Regarding the evaluation metrics, we utilize accuracy as the primary measure. Accuracy (A) is a widely used measure for evaluating the overall performance of a classification model. Specifically, it is calculated by determining the proportion of correct predictions, $TP + TN$, over the total number of predictions, $TP + TN + FP + FN$, where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. The specific calculation for accuracy is detailed in Eq. (4).

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Recall (R) measures the ability of the model to identify all relevant instances of a specific class, indicating the proportion of true positives that were correctly identified. In the context of attack classification, recall signifies the model's capability to detect malicious samples of a specific attack accurately. It is computed as:

$$R = \frac{TP}{TP + FN} \quad (5)$$

High recall values indicate that the attack detection model effectively identifies most of the malicious samples present in the dataset.

On the other hand, precision (P) is a measure that indicates how well a model avoids false positive predictions. A higher precision score indicates a model having a lower rate of misclassifying negative instances as positive, making it more reliable in identifying true positive samples. It is calculated as:

$$P = \frac{TP}{TP + FP} \quad (6)$$

Higher precision values imply that the SAINT model has a lower rate of misclassifying benign samples as malicious.

Finally, the F1 score is a balanced measure combining recall and precision into a single value, comprehensively evaluating the model's performance. It is the harmonic mean of recall and precision and is calculated as:

$$F1 = 2 * \frac{P * R}{P + R} \quad (7)$$

F1-score considers both false positives and false negatives and provides a balanced assessment of the model's ability to classify malicious and benign samples accurately. A higher F1-score indicates a better overall performance of the attack detection model.

To assess the stability of our robust scheme across various attack offsets, we employ the IQR of the accuracy values as an additional metric. The IQR evaluates the variability of accuracies and is computed by subtracting the upper quartile (Q3) from the lower quartile (Q1). Q1 denotes the threshold below which 25% of the accuracy values lie, while Q3 represents the threshold below which 75% of the accuracy values lie. The detailed calculation for the IQR is shown in Eq. (8).

$$IQR = Q3 - Q1 \quad (8)$$

5.1.3. Configurations

Throughout our experiments, we leveraged the default parameters for all models with minor modifications. Specifically, we trained our robust decision trees with a maximum depth of 7. In the case of SAINT, default parameters were employed, involving 100 epochs of execution, and categorical and numeric feature lists were provided based on the dataset. The allocation of data between the training and testing sets was established at a ratio of 80% for training and 20% for testing. Additionally, the data division for SAINT was 65% for training, 15% for validation, and 20% for testing, respectively.

5.2. Results and discussion

5.2.1. Our detection performance

In this section, we present the experimental outcomes of the enhanced robust decision trees on the three datasets, as depicted in Figs. 8, 9, and 10. The improved robust decision trees exhibit superior mean accuracy levels under small attack offsets. The accuracy variance is also less severe than in our initial experiment, where the defense offset was manually selected.

Regarding the IoTID20 dataset, Fig. 8 illustrates the outstanding performance of robust decision trees, which achieved a higher mean accuracy while exhibiting lower variance under small attack offsets. Specifically, for attack offsets ranging from 0.1 to 0.4, the average accuracy remained consistently high at 92%. Additionally, the IQR was remarkably narrow at 6%, with the third quartile centered around 95% and the first quartile approximating 89%.

In terms of the CIC-IDS-2017 dataset, Fig. 9 showcases the enhanced performance of robust decision trees, as they achieve higher accuracy across a broader range of attack offsets while maintaining minimal variance. Notably, the improved robust decision trees sustain an impressive mean accuracy of approximately 99% for the first four attack offsets and approximately 89% for an attack offset of 0.5. Moreover, the IQR remains exceptionally narrow, averaging around 1% under these attack offsets. Even at an attack offset of 0.6, the decision trees exhibit acceptable performance in detecting attacks, with a mean accuracy of just below 78% and an IQR of only 12%.

Finally, Fig. 10 presents an auspicious outcome achieved by the robust decision trees with our statistical solution on the BOT-IoT dataset, showcasing significant improvements in the mean accuracy and the IQR. Specifically, for attack offsets ranging from 0.1 to 0.3, the mean accuracy impressively reaches approximately 99.998%, with the IQR predominantly at 0%. This result marks a substantial enhancement when compared to our preliminary experiment on all other attack offsets, as the mean accuracy increases approximately threefold, and the first quartile experiences a remarkable surge from roughly 0% to 50%.

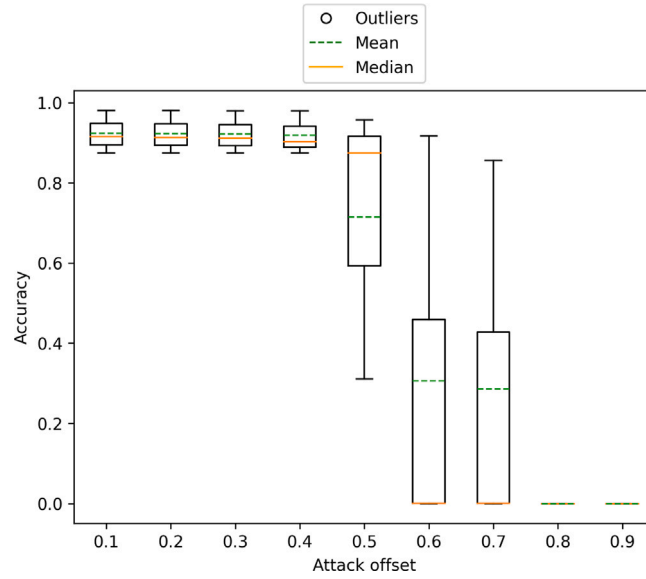


Fig. 8. Attack detection accuracy of the improved robust decision trees under different attack offsets on the IoTID20 dataset.

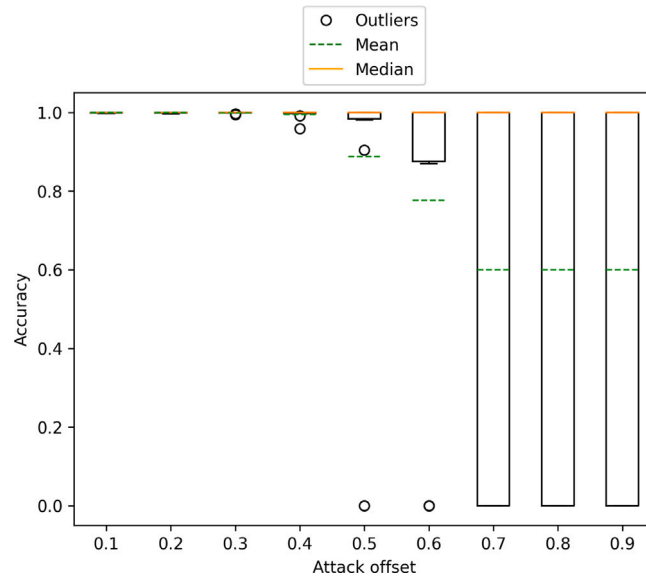


Fig. 9. Attack detection accuracy of the improved robust decision trees under different attack offsets on the CIC-IDS-2017 dataset.

5.2.2. Our attack classification performance

In this experiment, we aimed to assess the effectiveness of SAINT in accurately classifying various attack types of predictive malicious samples from the first detection stage. We utilized three commonly used evaluation metrics to ensure a fair evaluation across different classes: macro recall, macro precision, and macro F1-score.

The term *macro* in these metrics refers to the calculation done for each class independently and then averaging the results to obtain an overall score, resulting in an equally weighted metric for all classes regardless of class size. The comprehensive evaluation results can be accessed in Tables 5, 6, and 7. Our findings highlight SAINT's exceptional performance in classifying various attack categories across all three datasets, even under diverse and adversarial attack conditions. SAINT's remarkably high recall values on the CIC-IDS-2017 and BOT-IoT datasets are particularly noteworthy, achieving values of 99.065% and 98.3%, respectively. Furthermore, these two datasets exhibit high F1-scores of 97.925% and 96.902% and accuracy rates of 99.272% and 99.997%, respectively. SAINT's attack classification performance on the IoTID20 dataset is also decent, even in the face of adversarial attacks, as evidenced by the macro recall and macro F1-score values of 92.840% and 90.705%, respectively. Moreover, the accuracy of this dataset is also noteworthy, standing at 94.413%.

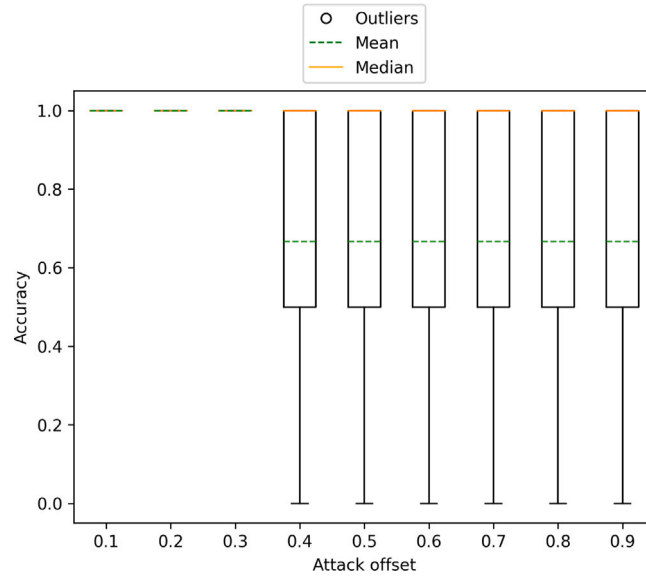


Fig. 10. Attack detection accuracy of the improved robust decision trees under different attack offsets on the BOT-IoT dataset.

Table 5

Macro recall, macro precision, macro F1-score, and accuracy of attack classification on the IoTID20 dataset.

Attack offset	Macro recall	Macro precision	Macro F1-score	Accuracy
0.1	0.9284	0.89452	0.90705	0.94413
0.2	0.9284	0.89454	0.90706	0.94413
0.3	0.9284	0.89454	0.90706	0.94413
0.4	0.9284	0.89454	0.90706	0.94413
0.5	0.9284	0.89454	0.90706	0.94413
0.6	0.9284	0.89455	0.90707	0.94413
0.7	0.92837	0.89455	0.90705	0.94413
0.8	0.92837	0.89455	0.90705	0.94413
0.9	0.92837	0.89455	0.90705	0.94413

Table 6

Macro recall, macro precision, macro F1-score, and accuracy of attack classification on the CIC-IDS-2017 dataset.

Attack offset	Macro recall	Macro precision	Macro F1-score	Accuracy
0.1	0.99065	0.96833	0.97926	0.99272
0.2	0.99065	0.96833	0.97926	0.99272
0.3	0.99065	0.96833	0.97926	0.99272
0.4	0.99065	0.96832	0.97925	0.99272
0.5	0.99063	0.96832	0.97924	0.99271
0.6	0.99063	0.96832	0.97924	0.99271
0.7	0.99063	0.96832	0.97924	0.99271
0.8	0.99063	0.96831	0.97924	0.99271
0.9	0.99063	0.96831	0.97924	0.99271

5.2.3. Discussion

Based on our experimental findings, it is evident that our enhanced robust decision trees exhibit good performance on small attack offsets, indicating their robustness to adversarial perturbations. Nonetheless, as the attack offset increases, the model's accuracy decreases, making it vulnerable to high-offset attacks. However, it is noteworthy that adversarial perturbations should be small enough to be both imperceptible and functional. An adversarial attack with a high offset would be easily noticeable and significantly distort the original traffic. Therefore, the ability of our model to maintain decent accuracy and stability on small attack offsets is a promising and acceptable performance for real-world scenarios.

Table 7

Macro recall, macro precision, macro F1-score, and accuracy of attack classification on the BOT-IoT dataset.

Attack offset	Macro recall	Macro precision	Macro F1-score	Accuracy
0.1	0.983	0.9561	0.96903	0.99997
0.2	0.983	0.9561	0.96903	0.99997
0.3	0.983	0.9561	0.96903	0.99997
0.4	0.983	0.9561	0.96903	0.99997
0.5	0.983	0.9561	0.96903	0.99996
0.6	0.98298	0.95609	0.96901	0.99992
0.7	0.98298	0.95609	0.96901	0.99992
0.8	0.98298	0.95609	0.96901	0.99992
0.9	0.98298	0.95609	0.96901	0.99992

Table 8

Comparing the proposed robust scheme with existing related works.

Dataset	Work	Detection model	Detection accuracy (%)	Classification accuracy (%)
IoTID20	[8]	Multistaged decision trees	99.98	99.88
	[51]	Random forest	98	83
	[52]	Single hidden layer feed-forward neural networks	93.51	N/A
	[53]	Long short-term memory	N/A	86.2
	Our work	Multistaged robust decision trees - SAINT	92.331	94.413
CIC-IDS-2017	[8]	Multistaged decision trees	99.99	99.97
	[54]	Deep neural networks	99.93	N/A
	[55]	Rules and decision trees	N/A	96.665
	[56]	Ensemble learning	N/A	99.11
	Our work	Multistaged robust decision trees - SAINT	99.944	99.272
BOT-IoT	[8]	Multistaged decision trees	99.99	99.99
	[55]	Rules and decision trees	N/A	96.996
	[57]	Long short-term memory	99.48	96.3
	[32]	Deep convolutional neural networks	N/A	98.86
	Our work	Multistaged robust decision trees - SAINT	99.998	99.997

5.3. Detection performance: Baseline comparison

To further evaluate the competitiveness of our robust scheme, we conducted a baseline comparison with other works, as presented in Table 8. We used the highest mean accuracy that our improved robust decision trees can achieve for attack detection across all adversarial attack offsets. The comparison revealed that our robust scheme performs on par with these competitors in terms of both attack detection and attack classification. Specifically, our competitors attained the best detection accuracy at 99.99%, while our best accuracy was 99.998%. Regarding classification performance, our competitors achieved the best results at 99.88%, 99.97%, and 99.99% on IoTID20, CIC-IDS-2017, and BOT-IoT, respectively. On the other hand, our model's performance is comparable and not far behind, with an accuracy of 94.932%, 99.111%, and 99.996% on the three datasets. This baseline comparison demonstrates that our proposal, trained under adversarial conditions, can still deliver competitive attack detection and classification.

6. Conclusion

This paper investigates the sensitivity of the state-of-the-art robust decision tree to the offset parameter. It reports a significant variance in its performance across nine different attack offsets. To address this issue, we propose a mitigation method that utilizes the input data distribution to automatically determine a suitable defense offset for the tree, resulting in improved mean accuracy and reduced variance. Additionally, we introduce a robust scheme for IDSs in IoT environments that incorporates our improved robust decision trees with SAINT, a tabular deep learning model. Our experimental results demonstrate that our IDS can deliver reliable attack detection and classification under diverse adversarial attack offsets.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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